🏠 Builder Basics — The Sloped Roof (Extension · Intermediate)

Topic: Variables · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Intermediate · **Time:** about 38 minutes

> 🧩 This is a bonus challenge, not a weekly lesson. Try it once you are comfortable building square rings with loops and you want to make a roof that steps up to a point.

A real roof is not one flat row — it is a stack of **square rings** that shrink as they go up. In the Sloped Roof world, typing `slope` builds the whole roof for you. This worksheet is about the **idea**, not typing code.

The program stores the ring's side length in a **variable** called `roof_len`. It starts at **7**, so the bottom ring is 7 stairs per side. After each ring it steps **up and inward**, then shrinks the variable by 2. Four rings get built, with sides of **7, 5, 3, 1**. Change the starting number and the whole roof changes shape.

1 · Predict 🔮

The variable `roof_len` starts at **7**. Each layer, the agent walks a square ring (4 sides, each `roof_len` stairs long), then shrinks `roof_len` by 2. The four rings have sides of **7, 5, 3, 1**.

What shape do the rings make as they stack up? Describe it in one sentence.

About how many stairs are placed in total across all four rings? Each ring's edge is about 4 × its side length, and the side lengths are 7, 5, 3, 1. Write the four numbers you got, then add them for the total. Show your working — do not just guess.

2 · Trace the Variable 🔢

Walk through the build one ring at a time. For each layer, write `roof_len` **for that ring**, then its value **after** it loses 2. Show the subtraction.

Layer	`roof_len` for this ring	`roof_len` after it loses 2
1		
2		
3		
4		

After the 4th ring, what is the value of `roof_len`? Why does the roof stop here instead of building one more ring?

3 · Reason About the Variable 🧠

No code here — think it through.

The roof comes out FLAT, like a box, instead of sloped. Every ring ended up the same size. Which part of the rule about the variable must have gone wrong for that to happen? Explain in one or two sentences.

Now imagine the variable lost 1 **each ring instead of** 2**. How would the roof look different — would the sides be steeper or gentler? Would there be more rings or fewer before it reached the peak? Explain why.**

4 · Make It Yours ✏️

The starting value of `roof_len` decides how wide the bottom ring is. Because the variable drops by 2 each ring, a bigger starting number takes more rings to shrink all the way down to 1.

Choose a starting `roof_len` so the roof has exactly 5 rings **and still shrinks neatly to a peak of 1. What starting value works? List the 5 ring sizes in order, from the widest bottom ring up to the peak.**

One sentence: how did you pick that starting value?

5 · Show Your Work 📸 🎥

Open the **Sloped Roof** world. Type `slope` in the chat to run the build and watch the rings shrink and step up to the peak. Then try the wider **5-ring** version by changing the starting number to the value you chose in Section 4, and run it again.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.